Anubhav **Jangra**

Computer Science Ph.D. student, Columbia University

Homepage @ anubhav@cs.columbia.edu Github Google Scholar

Education

Present – Aug 2023 | **Columbia University** | **GPA: 3.86/4.0**
Doctor of Philosophy in Computer Science (Advisor: Smaranda Muresan)

Jun 2021 – Jul 2017 | **Indian Institute of Technology, Patna** | **GPA: 8.82/10**
Bachelor of Technology in Computer Science and Engineering

Research Experience

Aug 2026 – May 2026 | **Microsoft Copilot Studio Team** [🌐] | **Redmond, USA**
Applied Science Intern | Samarth Gupta, Deepak Sharma
Designed and built a unified skill evaluation framework that quantifies how much a skill improves an autonomous agent across five interpretable dimensions. Evaluation was validated across multiple agents using SkillsBench, mutation testing to assess metric sensitivity, and a synthetic data pipeline grounded in real user needs that reduces reliance on hand-written verifiers. Adopted as the measurement backbone for the team's skill-authoring and skill-recommendation workflows.

Aug 2024 – May 2024 | **Microsoft Research & Office of Applied Research** | **Redmond, USA**
JEM (Joint E+D & MSR) Research Intern | Sujay Jauhar, Adrian de Wynter, Bahar Sarrafzadeh
Incorporated style personalized text generation (SPTG) into Word CoPilot's draft and rewrite text generation features to reflect users' authentic voices. Successfully launched the prototype for internal dogfooding. Investigated automatic evaluation metrics for low-resource SPTG across multiple writing domains and evaluation settings [*under submission at ARR*].

Jul 2023 – Jul 2021 | **Google Research | Advertising Sciences Team** [🌐] | **Bangalore, India**
Pre-Doctoral Researcher | Aravindan Raghuveer
Explored NLG techniques for creative advertisement generation. Investigating several research areas like text style transfer, data-to-text generation, automatic code generation, semantic representations etc.

Publications

US=under submission, P=Preprints, C=Conference, B=Book, SP=Short Paper, J=Journal

Selected Publications

[P] **`DivAlgo`: A systematic evaluation of algorithmic diversity for code generation models** [🔗]
Anubhav Jangra, Smaranda Muresan
arxiv 2610.19157 **[ArXiV'26]**

[P] **Fantastic Metrics and How to Find Them: Meta-Evaluating Metrics for Multi-lingual SPTG** [🔗]
Anubhav Jangra, Adrian de Wynter
arxiv 2610.19166 **[ArXiV'26]**

[P] **`CREATE`: Creativity Evaluation of VLMs using the Visual Remote Associates Test** [🔗]
Ziqing Yang*, Anubhav Jangra*, Smaranda Muresan
arxiv 2610.18569 **[ArXiV'26]**

[C] **Evaluating Style-Personalized Text Generation: Challenges and Directions** [🔗]
Anubhav Jangra, Bahareh Sarrafzadeh, Silviu Cucerzan, Adrian de Wynter*, Sujay Kumar Jauhar*
The 2026 Conference on Empirical Methods in Natural Language Processing, Budapest, Hungary (main) **[EMNLP'26]**

[P] **Designing and Evaluating Hint Generation Systems for Science Education** [🔗]
Anubhav Jangra, Smaranda Muresan
arxiv 2510.21087 **[ArXiv, 2025]**

[P] **Navigating the Landscape of Hint Generation Research: From the Past to the Future** [🔗]
Anubhav Jangra, Jamshid Mozafari, Adam Jatowt, Smaranda Muresan
Transactions of the Association for Computational Linguistics (presented at ACL 2025, Vienna, Austria) **[TACL'25]**

[C] **Large Scale Multi-modal Multi-lingual Summarization Dataset** [🔗]
Yash Verma*, Anubhav Jangra*, Raghvendra Verma, Sriparna Saha
The 17th Conference of the European Chapter of the Association for Computational Linguistics, Dubrovnik, Croatia **[EACL'23]**

[C] **T-STAR: Truthful Style Transfer using AMR Graph as Intermediate Representation** [🔗]
Anubhav Jangra*, Preksha Nema*, Aravindan Raghuveer
The 2022 Conference on Empirical Methods in Natural Language Processing, Abu Dhabi, UAE **[EMNLP'22]**

[J] **A Survey on Multi-modal Summarization** [🔗]
Anubhav Jangra, Sourajit Mukherjee, Adam Jatowt, Sriparna Saha, Mohammed Hasanuzzaman,
ACM Computing Surveys **[ACM CSUR'23]**

[C] **WIDAR - Weighted Input Document Augmented ROUGE** [🔗]
Raghav Jain*, Vaibhav Mavi*, Anubhav Jangra*, Sriparna Saha
44th European Conference on Information Retrieval, Stavanger, Norway **[ECIR'22]**

[C] **Multi-modal Supplementary Complementary Summarization using Multi-Objective Optimization** [🔗]
Anubhav Jangra, Sriparna Saha, Adam Jatowt, Mohammed Hasanuzzaman
44th International ACM SIGIR Conference on Research and Development in Information Retrieval, Virtual **[SIGIR'21]**

[C] **Semantic Extractor Paraphraser based Abstractive Summarization** [🔗]
Anubhav Jangra*, Raghav Jain*, Vaibhav Mavi*, Sriparna Saha, Pushpak Bhattacharyya,
17th International Conference on Natural Language Processing, Patna, India **[ICON'20]**

[SP] **Multi-Modal Summary Generation using Multi-objective Optimization** [🔗]
Anubhav Jangra, Sriparna Saha, Adam Jatowt, Mohammed Hasanuzzman,
43rd International ACM SIGIR Conference on Research and Development in Information Retrieval, Xi'an, China **[SIGIR'20]**

[SP] **Text-Image-Video Summary Generation using Joint Integer Linear Programming** [🔗]
Anubhav Jangra, Adam Jatowt, Mohammed Hasanuzzman, Sriparna Saha,
42nd European Conference on Information Retrieval, Lisbon, Portugal **[ECIR'20]**

Collaborative Publications

[P] **Evidence-Anchored Overdose Death Prediction from Death Investigation Reports** [🔗]
Raavi Gupta, Ziqing Yang, Karli R. Hochstatter, Talia Nadel, Nicole D'Anna, Jason K. Graham3, Hannah Johnson, Kelsey Opozda, Nabila El-Bassel, Anubhav Jangra, Smaranda Muresan
arXiv 2610.19556 **[ArXiv, 2026]**

[J] **Human-centered artificial intelligence framework for overdose mortality surveillance** [🔗]
Karli R Hochstatter, Talia Nadel, Raavi Gupta, Anubhav Jangra, Nicole D'Anna, Jan Gryczynski, Yuxin Li, Jason Graham, Hannah Johnson, Nabila El-Bassel, Ziqing Yang, Smaranda Muresan
American Journal of Public Health, 2026 **[AJPH'26]**

[P] **LLMs as Science Journalists: Supporting Researchers in Communicating Their Science to the Public** [🔗]
Milad Alshomary, Grace Li, Anubhav Jangra, Yufang Hou, Kathleen McKeown, Smaranda Muresan
arXiv 2601.05821 **[ArXiv, 2026]**

[B] **Multi-hop Question Answering** [🔗]
Vaibhav Mavi, Anubhav Jangra, Adam Jatowt
Foundations and Trends® in Information Retrieval Vol. 17 Issue 5 **[FnTs, 2024]**

[C] **TriviaHG: A Dataset for Automatic Hint Generation for Factoid Questions**
Jamshid Mozafari, Anubhav Jangra, Adam Jatowt
The 47th International ACM SIGIR Conference on Research and Development in Information Retrieval, Wash. D.C., USA **[SIGIR'24]**

[C] **Can Multimodal Pointer Generator Transformers produce topically relevant summaries?** [🔗]
Sourajit Mukherjee, Anubhav Jangra, Sriparna Saha, Adam Jatowt,
2023 International Joint Conference on Neural Networks (IJCNN) **[IJCNN'23]**

[C] **Topic-aware Multimodal Summarization** [🔗]
Sourajit Mukherjee, Anubhav Jangra, Sriparna Saha, Adam Jatowt,
2nd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics **[Findings in AACL'22]**

[P] **A Survey on Medical Document Summarization** [🔗]
Raghav Jain, Anubhav Jangra, Adam Jatowt, Sriparna Saha
ArXiv 2212.01669 **[ArXiv, 2022]**

[C] **Combining Vision Language Representations for Patch-based Identification of Lexico-Semantic Relations** [🔗]
Prince Jha, Gaël Dias, Alexis Lechervy, José G Moreno, Anubhav Jangra, Sebastião Pais, Sriparna Saha
30th ACM International Conference on Multimedia, Lisbon, Portugal **[ACM MM'22]**

[C] **MAKED: Multi-lingual Automatic Keyword Extraction Dataset** [🔗]
Yash Verma, Anubhav Jangra, Sriparna Saha, Adam Jatowt, Dwaipayan Roy
13th Conference on Language Resources and Evaluation **[LREC'22]**

[J] **Identifying Complaints based on Semi-Supervised Mincuts** [🔗]
Apoorva Singh, Sriparna Saha, Mohammed Hasanuzzaman, Anubhav Jangra
Elsevier's Expert Systems with Applications, Volume 186, 2021 **[ESWA'21]**

[J] **Extractive Single Document Summarization using Multiobjective Optimization: Exploring Self-organized Differential Evolution, Grey Wolf Optimizer and Water Cycle Algorithm** [🔗]
Naveen Saini, Sriparna Saha, Anubhav Jangra, Pushpak Bhattacharyya,
Elsevier's Knowledge Based Systems, 2018 **[KBS'18]**

Other Experiences

Aug 2020 – Jul 2020 | **GREYC Lab, ENSI-CAEN** [🌐] **Remote / Caen, France**
Research Intern | Supervisor: Gaël Dias
Extended patch-based lexical semantic identification frameworks to a multi-modal setting. Developed the dataset and conducted the pilot studies of the project. [*ACM MM'22*]

Jun 2019 – May 2019 | **Graduate School of Informatics, Kyoto University** [🌐] **Kyoto, Japan**
Research Intern | Supervisor: Adam Jatowt
Explored various unsupervised optimization techniques to develop multi-modal summarization systems that generate text-image-audio-video summaries.

Jul 2021 – Jun 2021 | **IBM** **Remote / Chennai, India**
Global Research Mentee | Supervisor: Ganesan Narayanasamy
Developed the project framework for *Health Care App*, that uses knowledge graphs and named-entity recognition to help users self-diagnose themselves.

Mar 2021 – Dec 2020 | **Huawei Technologies Co., Ltd** **Remote**
Project Member | Supervisor: Sriparna Saha
Developed a Proof of Concept (POC) for the task of automatic tagline generation and product description using existing neural summarization systems for the upcoming collaborative project of IIT Patna and Huawei.

Jan 2020 – Dec 2019 | **TCS Innovation Lab** [🌐] **Kolkata, India**
Research Intern | Supervisor: Arijit Ukil
Investigated generative modeling to tackle the insufficiency of data in time-series signal classification.

Jan 2019 – Dec 2018 | **CFILT Lab, IIT Bombay** [🌐] **Mumbai, India**
Research Intern | Supervisor: Pushpak Bhattacharyya
Examined Unsupervised NMT for distant language pairs (Indo-Aryan languages) using attention based seq2seq models.

Jul 2021 – Jul 2018 | **AI-NLP-ML Lab, IIT Patna** [🌐] **Patna, India**
Undergraduate Research Scholar | Supervisor: Sriparna Saha
Worked extensively in the area of summarization (*e.g.,* multi-modal summarization, extractive and abstractive text summarization), complaint mining and multi-label classification.

Academic Service

PC Member Coling 2025, LREC-Coling 2024, Text2Story Workshop (ECIR 2023, 2024), IACT (SIGIR 2023)

Reviewer (Conference) ACL ARR (*since Dec 2023*), CIKM 2023, ACL 2023, EMNLP 2022

Reviewer (Journals) ACM Computing Surveys (*since Jan 2021*), ACM TALLIP (*since May 2020*), Applied Artificial Intelligence (*since Oct 2021*), and IEEE Transactions on Computational Social Systems (*since Jan 2022*), Expert Systems with Applications (*since Sept 2022*), Engineering Applications of Artificial Intelligence (*since Feb 2022*), IEEE Internet Computing (*since Feb 2022*)

Secondary Reviewer AAAI 2020, EACL 2021, ACL 2021, EMNLP 2021, CIKM 2021, KDD 2022, and WebConf 2022

Research Mentor Mentoring two masters and two undergraduate student towards building intelligent tutoring systems at Columbia University. Mentored three undergraduate interns, two masters student researchers, and one undergraduate students for automatic summarization at IITP.

Community Service Organizer of CARE program—a Community for AI Research and Education to guide students and early-stage researchers through their research-related queries.

Volunteer Volunteered as a reviewer in Google's CS Research Mentorship program to help review applicants from historically marginalized communities for the mentorship program. Reviewer for PhD Pre-Application Review (PAR) program at Columbia university.

Talks

- Seminar Talk: **Scientific Hint Generation** - Learning with and about AI at Carnegie Mellon University — Feb 2026
- Seminar Talk: **Scientific Hint Generation** - NLP Seminar at Columbia University — Oct 2025
- Guest lecture: **Hands-on Deep Learning** - COMSW4705 (NLP by Daniel Bauer), Columbia University — Oct 2025
- Guest lectures: **Intro to Deep Learning** - COMSBC3705 (NLP by Smaranda Muresan), Columbia University — Sept 2025
- **Abstract Meaning Representation (AMR) Graphs at work!** - AI-NLP-ML Lab, IIT Patna — Oct 2022
- **Automatic Text Summarization** PyData Patna Conference — Dec 2020

Teaching and Leadership Roles

Barnard College at Columbia University *Head Teaching Assistant* Jan 2026-May 2026

- Head teaching assistant for BC3707 (LLMs Foundations and Ethics) course taught by Dr. Smaranda Muresan.

Department of Computer Science at Columbia University *Head Teaching Assistant* Sep 2025-Dec 2025

- Head teaching assistant for COMSW4705 (Natural Language Processing) course taught by Dr. Daniel Bauer.

Barnard College at Columbia University *Teaching Assistant* Sep 2024-Dec 2024

- Teaching assistant for BC3997 (Natural Language Processing) course taught by Dr. Smaranda Muresan.

Google DSC IIT Patna, Patna, India *ML Department Lead* 2019-2020

- Supervised three projects and gave lectures on Machine Learning theory and its applications.

Univeristy of Innsbruck, Austria *Teaching Assistant* Jun 2020

- Part-time Teaching Assistant in the course 2021S703836 VU (Natural Language Processing). Prepared lectures on automatic summarization.